

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457887.67321 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.085 +/- 0.035
Transit depth (Rp/Rs)^2	0.72 +/- 0.59 [%]
Orbital Inclination (inc)	83.3 +/- 2.78
Airmass coefficient 1 (a1)	1.203 +/- 0.015
Airmass coefficient 2 (a2)	-0.1195 +/- 0.0054
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	None
Optimal Aperture	7.63
Optimal Annulus	8.9
Transit Duration (day)	0.052 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.085 $\pm$ 0.035 vs 0.1214 (deviation 0.99 σ)
Duration fit vs published	0.052 d vs 0.0742 d (deviation 0.81 σ)
Mid-time O-C	-1.6 min
Depth SNR	2.3
Residual scatter	1.22 %

Light curve

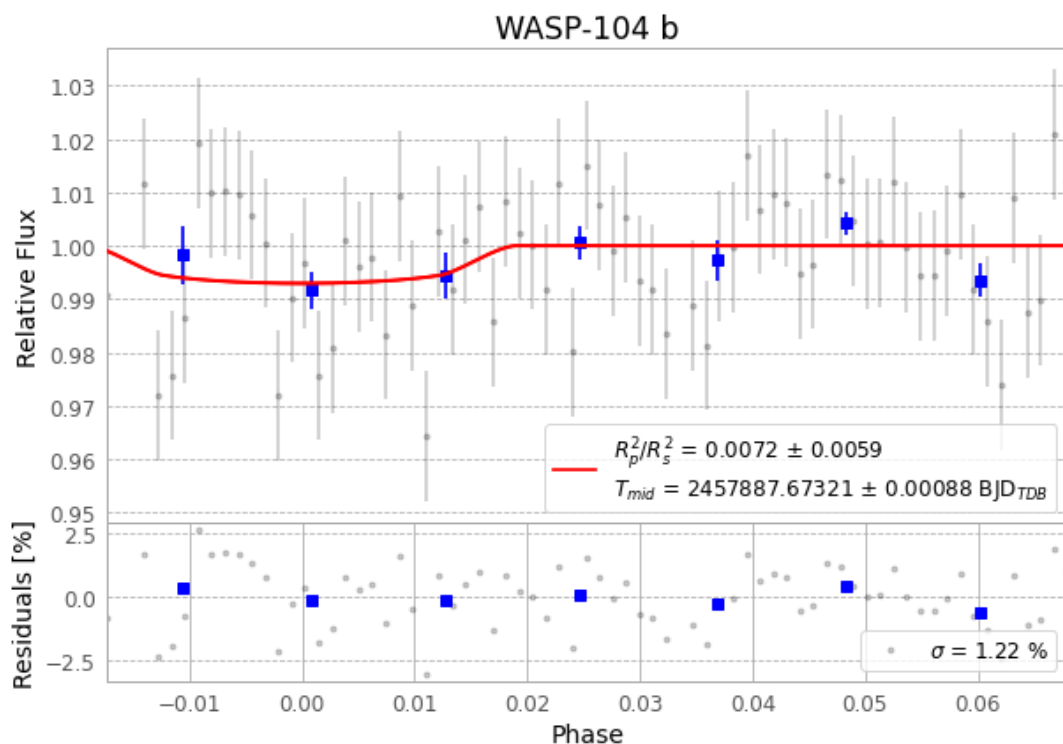


Figure 1 — FinalLightCurve_WASP-104 b_2017-05-13.png (SHA-256 e84060396025494c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0636c0c5a97f559e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), WASP-104170514032143.FITS → WASP-104170514065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-170514.log (7577c737d808...), FinalLightCurve_WASP-104 b_2017-05-13.png (e84060396025...), FinalLightCurve_WASP-104 b_2017-05-13.csv (994c7577267a...)

Compute resources

Wall time 133.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 01:27Z.